

matsuo-lab / **weblab-10b**   like 63

 Text Generation  Transformers  PyTorch gpt_neox
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  Train  Deploy  Use this model

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 Text Generation

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 **weblab-10b**

 **Overview**

This repository provides a Japanese-centric multilingual GPT-NeoX model of 10 billion parameters.

- **Library**

The model was trained using code based on [EleutherAI/gpt-neox](#).

- **Model architecture**

A 36-layer, 4864-hidden-size transformer-based language model.

- **Pre-training**

The model was trained on around **600B** tokens from a mixture of the following corpora.

- [Japanese C4](#)
- [The Pile](#)
- **Model Series**

Variant	Link
weblab-10b-instruction-sft	https://huggingface.co/matsuo-lab/weblab-10b-instruction-sft
weblab-10b	https://huggingface.co/matsuo-lab/weblab-10b

- **Authors**

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[🔗](#) Benchmarking

- **Japanese benchmark : JGLUE 8-task (2023-08-27)**
 - We used [Stability-AI/lm-evaluation-harness](#) library for evaluation.
 - The 8-task average accuracy is based on results of JCommonsenseQA-1.1, JNLI-1.1, MARC-ja-1.1, JSQuAD-1.1, jaqket_v2-0.2, xlsun_ja-1.0, xwinograd_ja, and mgsm-1.0.
 - model loading is performed with float16, and evaluation is performed with template version 0.3 using the few-shot in-context learning.
 - The number of few-shots is 3,3,3,2,1,1,0,5.
 - `special_tokens_map.json` is modified to avoid errors during the evaluation of the second half benchmarks. As a result, the results of the first half benchmarks became slightly different.

model	average	jcommonsenseqa	jnli	marc_ja	jsquad	jaqket_v2	xlsum_ja	xwinog
weblab-10b-instruction-sft	59.11	74.62	66.56	95.49	78.34	63.32	20.57	71.95
weblab-10b	50.74	66.58	53.74	82.07	62.94	56.19	10.03	71.95

- **Japanese benchmark : JGLUE 4-task (2023-08-18)**

- We used [Stability-AI/lm-evaluation-harness](#) library for evaluation.
- The 4-task average accuracy is based on results of JCommonsenseQA-1.1, JNLI-1.1, MARC-ja-1.1, and JSQuAD-1.1.
- model loading is performed with float16, and evaluation is performed with template version 0.3 using the few-shot in-context learning.
- The number of few-shots is 3,3,3,2.

Model	Average	JCommonsenseQA	JNLI	MARC-ja	JSQuAD
weblab-10b-instruction-sft	78.78	74.35	65.65	96.06	79.04
weblab-10b	66.38	65.86	54.19	84.49	60.98

🔗 How to use the model

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
```

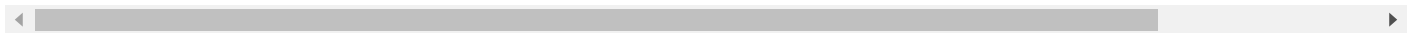
```
tokenizer = AutoTokenizer.from_pretrained("matsuo-lab/weblab-10b")
model = AutoModelForCausalLM.from_pretrained("matsuo-lab/weblab-10b", torch_dtype=

if torch.cuda.is_available():
    model = model.to("cuda")

text = "吾輩は猫である。"
token_ids = tokenizer.encode(text, add_special_tokens=False, return_tensors="pt")

with torch.no_grad():
    output_ids = model.generate(
        token_ids.to(model.device),
        max_new_tokens=100,
        do_sample=True,
        temperature=0.7,
        top_p=0.95
    )

output = tokenizer.decode(output_ids.tolist()[0])
print(output)
```



[🔗](#) **Licenses**

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